

OfferPrep Development Roadmap (v1)

Goal

Build OfferPrep as a production-quality Full Stack + AI + Machine Learning project, completing one stable phase at a time without leaving unfinished work.

Phase 1 – Authentication (Completed)

- React + Vite + Tailwind setup
- FastAPI backend
- PostgreSQL (Neon) + SQLAlchemy + Alembic
- JWT Authentication
- Register/Login APIs
- Protected Routes
- Dashboard
- Logout
- README
- QA & Cleanup

Phase 2 – Resume Management

- Resume Upload UI
- Resume Upload API
- Store PDF securely
- Resume database table
- Replace/Delete Resume
- Resume Dashboard

Phase 3 – Resume Parsing (NLP)

- Extract text from PDF
- Extract Skills
- Extract Education
- Extract Experience
- Extract Projects
- Store parsed data

Phase 4 – AI Interview

- Gemini integration
- Generate interview questions
- Technical/HR interviews
- Difficulty levels

- Company-specific interviews

Phase 5 – Machine Learning

- Create/collect dataset
- Feature engineering
- Train ML model (Random Forest recommended)
- Resume Score Prediction
- Placement Readiness Prediction
- Skill Gap Analysis
- Explain predictions

Phase 6 – AI Feedback

- Evaluate answers
- Score responses
- Improvement suggestions
- Overall interview feedback

Phase 7 – Analytics

- Interview history
- Performance charts
- Skill progress
- Weak/strong areas

Phase 8 – Deployment

- Deploy frontend
- Deploy backend
- Production database
- Environment configuration
- Domain & testing

Development Workflow

For every feature: Plan → Backend → Frontend → Testing → Git Commit → README Update

Milestone Goal

Complete each phase fully before starting the next. Avoid unfinished features. Maintain production-quality code, documentation, and Git history.